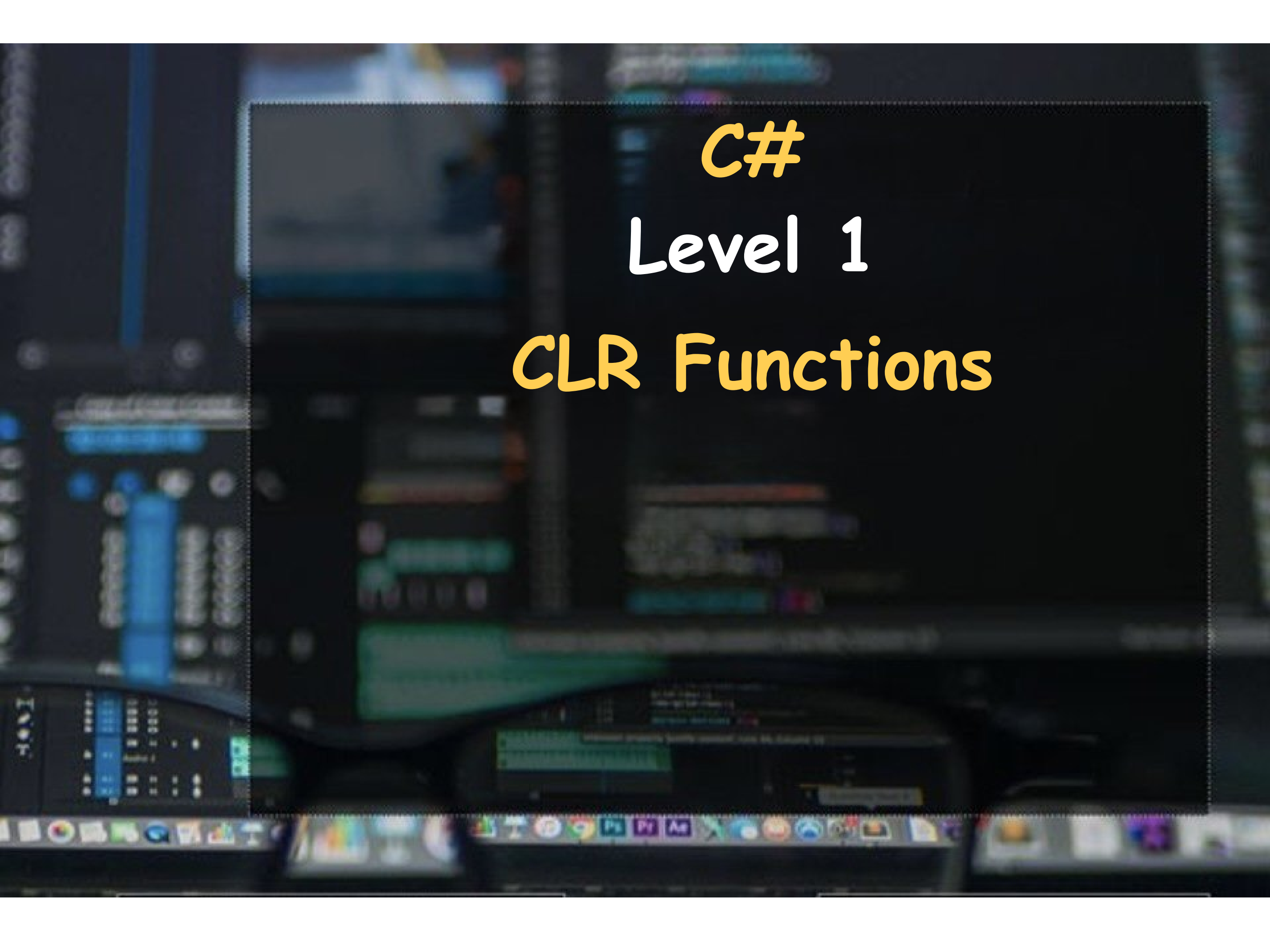
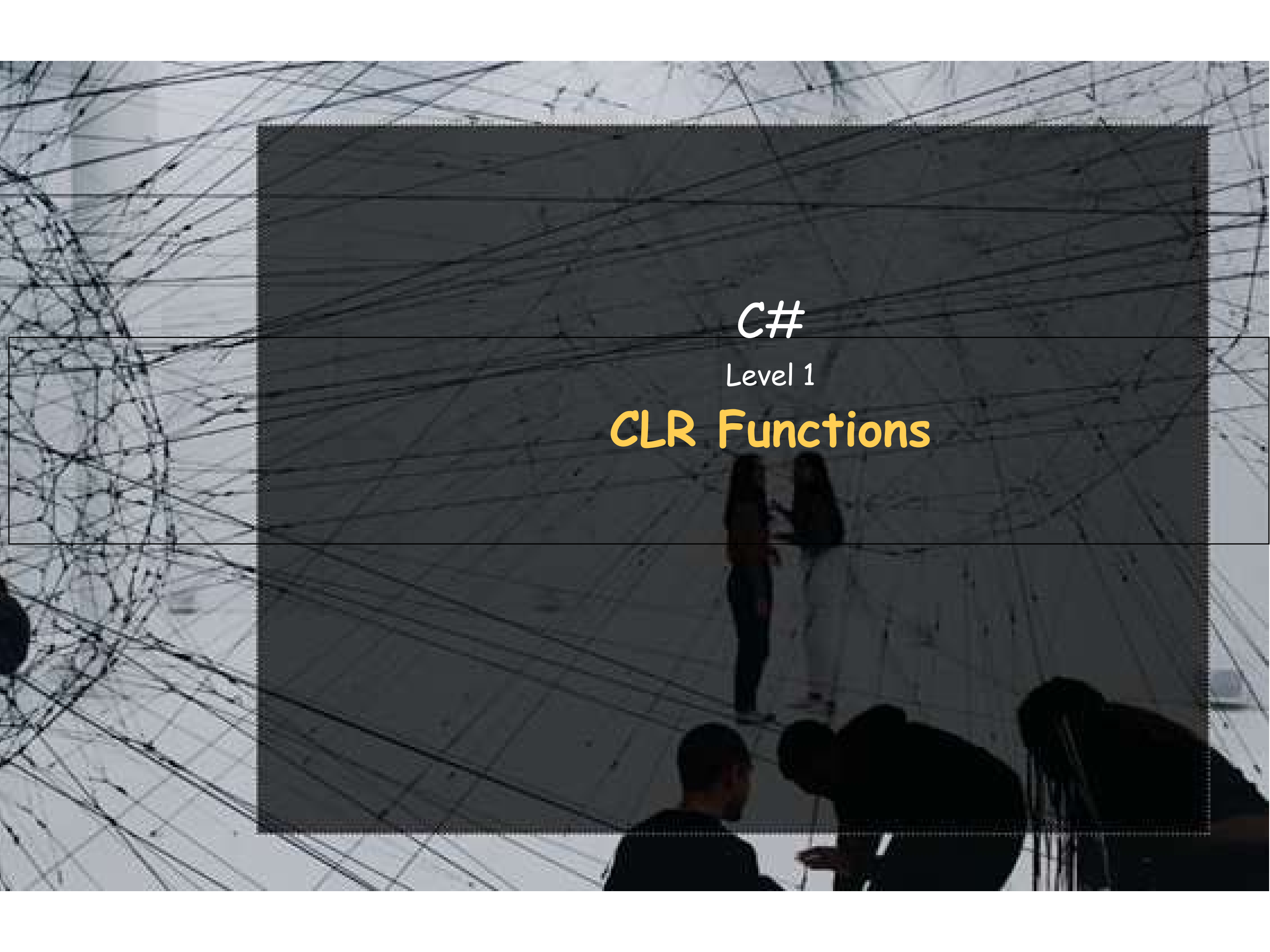


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, emphasizing the text overlay.

C# Level 1 CLR Functions



C#

Level 1

CLR Functions

Definition

CLR

-  CLR (Common Language Runtime) is the execution environment of .NET.
-  It manages and supports the execution of .NET applications.

Explanation

Main CLR Responsibilities

-  JIT compilation.
-  Garbage collection.
-  Exception handling.
-  Type safety.
-  Managed memory management.
-  Runtime execution services.

Language and Platform Support

-  Multiple .NET languages can target the same runtime.
-  Modern .NET supports execution on Windows, Linux, and macOS.

Procedures

1. CLR loads the application.
2. CLR provides runtime services.
3. JIT compiles required IL into native code.
4. CLR manages memory and exceptions.
5. Application continues execution under the runtime environment.

💡 Example

.NET Application



CLR



JIT + GC + Runtime Services



Operating System



Hardware

➡ Result

The CLR manages important runtime tasks while the .NET application executes.

Note

- ◆ JIT is a component involved in CLR-based execution, not the CLR itself.
- ◆ Modern .NET is cross-platform; old .NET Framework is primarily Windows-only.

⊘ Warning

- ◆ Do not memorize "CLR = JIT".
- ◆ JIT is responsible for IL-to-native compilation, while CLR provides many additional runtime services.



Comparison



CLR

- ✓ Manages .NET application execution.
- ✓ Provides runtime services.
- ✓ Works with JIT and Garbage Collection.
- ✓ Supports type and exception handling.



Operating System

- ✓ Manages system resources.
- ✓ Manages processes and hardware.
- ✓ Provides OS-level services.

✦ Summary

- ✓ CLR = Common Language Runtime.
- ✓ It is the .NET execution environment.
- ✓ Works with JIT.
- ✓ Provides Garbage Collection.
- ✓ Supports exception handling and type safety.
- ✓ Manages important runtime services.

Thank you 😊